WEEK 1 PROGRESS REPORT

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

University Student-Support Case Agent

Group M Evening | **Week Ending:** *4th Sept 2026*

Work Completed Against Weekly Objectives

All five Week 1 deliverables specified in the assignment brief were completed:

Selected and scoped the "University student-support case agent" use case from the approved list and defined a single primary end-to-end workflow (freshman asks an onboarding question > grounded answer or escalation > case creation/routing > human resolution).

Project Charter - problem, user, current pain point, AI value, scope, assumptions/constraints and a Minimum Proposal Statement.

User Stories and Acceptance Criteria - ten testable user stories (US-1 to US-10) covering grounded Q&A, honest fallback handling, timetable/calendar lookup, case creation, automatic categorisation and routing, session memory, escalation, iteration limits, adversarial/out-of-scope handling, and input validation.

AI Boundary Matrix - every meaningful agent action classified as AI-autonomous, deterministic/rule-based, or requiring human approval.

Initial architecture/context diagram - shows the agent's core components (LLM, RAG retrieval, agent decision engine, ticket-creator tool, state/memory management) and external entities (student, university documents, case database, support-desk staff, external LLM API, synthetic student records).

GitHub repository and ClickUp workspace created, with Week 1 tasks logged and tracked to completion.

Key Engineering Decisions

Narrowed scope to freshmen-only onboarding concerns (moved away from an initial plan to cover the entire Makerere information ecosystem for all student types). This aligns with the brief's requirement to solve "one clearly bounded problem for a defined user" rather than build a generic assistant and keeps the planned 10-50 document corpus realistic within eight weeks.

Adopted an external LLM API (OpenAI/Claude/Gemini-class model) rather than a self-hosted or fine-tuned model. The team's current strengths are in frontend/UI-UX and early AI exposure, so an API integration lets engineering effort go toward RAG, tools and agent bounding rather than model infrastructure.

Separated the Case Management Database from the RAG document corpus in the architecture. This keeps read-only, source-cited retrieval cleanly separated from auditable case/ticket writes, directly supporting the AI Boundary Matrix split between what the AI may draft and what deterministic software must record.

Made case-category routing AI-autonomous via a fixed category-to-queue lookup table, rather than requiring staff confirmation on every case. Routing is low-stakes and reversible and gating every case would undercut the efficiency gain the agent is meant to deliver; a human reassignment safety net is kept instead of a pre-approval gate.

Failures, Challenges and Current Response

ClickUp unfamiliarity: most team members were using ClickUp for the first time and initially struggled to navigate boards, statuses and task assignment. Response: Namata Sumayya, who set up the workspace, walked the team through it; a short ClickUp orientation is planned at the start of Week 2 to prevent repeated confusion.

Scope creep risk: the initial proposal attempted to cover the entire Makerere information ecosystem for all student types, which risked violating the brief's proposal rules and would have made the document corpus and timeline unrealistic. Response: the group deliberately re-scoped to freshmen-only onboarding concerns before finalising the Charter (see Key Engineering Decisions above).

Links and Evidence

GitHub repository (private, main branch): https://github.com/SumayyaN/BSE41014_capstone_group_m_eve

Latest commit: ac2edfd – "Upload AI Boundary Matrix", committed by Joshua-Aryampa (github.com/SumayyaN/BSE41014_capstone_group_m_eve/commit/ac2edfd). 5 commits recorded to date under docs/requirements; full history is in the repository's commit log.

ClickUp board: BSE4104_CAPSTONE_PROJECT_BY_GROUP_M_EVE - Week 1 board shows 4 tasks Complete (Project Charter, GitHub + ClickUp setup, User Stories and Acceptance Criteria, AI Boundary Matrix, Initial architecture/context diagram) and 11 tasks queued in To Do for upcoming weeks.

Individual Contribution Summary

Member	Task Owned	Evidence
Namata Sumayya	GitHub repository and ClickUp workspace setup; Project Charter	GitHub repo created under SumayyaN account; README.md commit; ClickUp tasks "Project Charter" and "GitHub + ClickUp setup" marked Complete
Aryampa Joshua	AI Boundary Matrix; Week 1 progress report	GitHub commit ac2edfd "Upload AI Boundary Matrix" (docs/requirements); this report. ClickUp tasks "AI boundary matrix" marked Complete
Nakaye Hildah	User Stories and Acceptance Criteria (US-1 to US-10)	ClickUp task "User stories and acceptance criteria" marked Complete; document committed to docs/requirements
Mushabe Moses	Initial architecture/context diagram	ClickUp task "Initial architecture/context diagram" completed; draft diagram file (final_initial_context_Architecture.drawio)

Plan for Next Week (Week 2 - Foundation-Model Engineering and Prompting)

a) Finalise and document model selection (Model Selection Note, 1 page maximum): capability, cost, latency, privacy and access considerations for the chosen LLM API.

b) Integrate the selected model into a working baseline application.

c) Draft Prompt Specification v1.0: role, task, context, constraints, output format and failure behaviour.

d) Build a 10-case prompt evaluation table recording expected vs. actual behaviour.

e) Version at least two meaningful prompt iterations.

f) Set up the Week 2 board.